Recovery and ROI model

The business case by program

Status: Draft v0.1 (2026-11-04), for Steve's review. Extension 3 of 10 (see the extensions overview). Model: `analytics/roi/model.py`; inputs: `analytics/roi/programs.csv`; results: `docs/proposals/data/roi_results.csv`. Interactive version on the site at `/tools/roi/`. **These are estimates for a business case, not savings. Savings stay Pending until an Inspector General or DOJ confirms them.**

Bottom line

Leaders fund what pays back. Starting from each program's reported improper payments (GAO, FY2025) and a stated chain of assumptions, the model estimates the five-year present value of what detection tooling would prevent or recover, against what it costs to build, run and review.

- **The large health and benefit programs pay back many times over even on cautious assumptions.** Medicare fee-for-service: median net benefit about $880M over five years, roughly 10 times cost; Medicaid about $450M (8 times), even with only a fifth of Medicaid reachable through state data.
- **The break-even bar is low.** For Medicare, the tooling pays for itself if it catches 0.2% of the improper dollars it can see that current controls miss. For SNAP it is 0.8%. A pilot only has to beat that number.
- **Small programs do not pay back on their own.** OPM retirement, FEMA Public Assistance and the closed Shuttered Venue grant program have a 50% or lower chance of positive net benefit as stand-alone builds. They should ride on shared screens (cross-agency matching, exclusion checks), not get their own tooling.

Results by program

Program	Improper	Benefit / cost	Net, P10 to P50	ROI	Break-even
Medicare FFS	$28.8B	$993M / $100M	$499M to $884M	9.7x	0.17%
Medicaid	$37.4B	$516M / $62M	$249M to $449M	8.0x	0.32%
EITC	$21.1B	$291M / $45M	$131M to $244M	6.4x	0.57%
SSI	$7.3B	$252M / $42M	$111M to $208M	5.9x	0.66%
SNAP	$10.2B	$211M / $38M	$89M to $171M	5.4x	0.79%
VA (7 programs)	$3.9B	$135M / $32M	$49M to $102M	4.2x	1.2%
HUD rental aid	$4.3B	$119M / $31M	$41M to $87M	3.8x	1.4%

Program	Improper	Benefit / cost	Net, P10 to P50	ROI	Break-even
UI	$5.6B	$116M / $31M	$39M to $85M	3.8x	1.4%
SSDI	$2.3B	$79M / $28M	$20M to $51M	2.9x	2.1%
SVOG (closed)	$10.1B	$32M / $31M	-$14M to $1M	1.0x	6.9%
OPM retirement	$0.4B	$20M / $23M	-$13M to -$3M	0.9x	8.3%
FEMA PA	$0.3B	$12M / $22M	-$18M to -$9M	0.6x	13%

Improper payments are FY2025. Benefit, cost and ROI are five-year medians; break-even is the catch rate at which net benefit is zero. Figures are present values at a 2% real discount rate, from 20,000 Monte Carlo draws. Run `python3 analytics/roi/model.py` to reproduce them.

1. The chain

For each program and year:

- **caught** = improper payments x reach x detectable x new detection x ramp
- **flagged** = caught / precision (what reviewers have to look at)
- **benefit** = caught x [prevent + (1 - prevent) x recovery] x (1 + deterrence)
- **cost** = build (year 1) + run (every year) + review cost x flagged

Closed programs get no prevention and no deterrence: only recovery on the stock of improper payments, spread over three years.

2. Assumptions and why

Each share is drawn from a triangular distribution (low, likely, high). These are the numbers to argue about with an agency, and the pilot is designed to measure the three that matter most (new detection, precision, prevent).

Assumption	Low	Likely	High	Basis
Reach	per program (0.2 to 1.0)			Share of program dollars our data access covers within five years. Medicaid and EITC 0.2 (state data, tax-data limits); Medicare 0.5; OPM 0.8
Detectable	25%	40%	55%	Share of improper dollars with a data trace (identity, eligibility, billing pattern) rather than a missing document. Agencies report most improper payments as documentation errors; the detectable share is what the screens can reach
New detection	3%	8%	15%	Share of detectable dollars our screens catch that current controls miss. The single most uncertain number; the pilot measures it
Precision	30%	50%	70%	Share of flagged dollars confirmed on review. Sets reviewer workload
Prevent	20%	45%	70%	Share caught before payment. Needs pre-payment integration (Do Not Pay, claims edits)



Assumption	Low	Likely	High	Basis
Recovery	5%	15%	30%	Share recovered when caught after payment. Pay-and-chase recovers little; GAO and OIG reports show low recovery rates
Deterrence	0%	5%	15%	Extra benefit from fewer attempts once controls are known
Build	$2M	$4M	$8M	One-time, per program, on the shared platform
Run	$1.5M	$3M	$6M	Per year, per program: data, compute, analysts
Review cost	1%	2%	4%	Review and investigation cost per flagged dollar
Ramp	25%, 60%, 100%, 100%, 100%			Capacity reached in years 1 to 5
Discount rate	2% real			OMB Circular A-94 (2023 revision) default; check the current value

3. What the model does not count

- Criminal fines, False Claims Act damages and penalties (often larger than the recovered overpayment; treble damages under 31 U.S.C. 3729).
- Fraud outside improper payment estimates. GAO estimated total fraud losses at $233B to $521B a year (GAO-24-105833), well above reported improper payments, so starting from improper payments is conservative.
- Shared-platform savings: the second and later programs reuse the entity spine and screens, so build cost per program falls.
- Benefits to states (the state share of Medicaid, UI administration).

4. Sensitivity

The ranking is driven by improper dollars x reach. Within a program, net benefit moves almost one for one with new detection and prevent. Precision matters through review cost only; at 30% precision review cost roughly doubles, but no large program turns negative. The break-even column is the best single number for a leader: if the pilot shows a catch rate above it, scale; if not, stop.

5. How to use it in a proposal

1. Quote the program's row, the break-even rate and the P10 (pessimistic) net, not the median.
2. Replace our assumptions with the agency's own where it has them (recovery rates, review cost).
3. State that savings stay Pending until confirmed, and that the pilot exists to measure the uncertain assumptions.

